Glossary

Aggregate interlock Shear or friction resistance by the concrete on opposite sides of a crack in a reinforced concrete member (obviously larger with narrower cracks).

Balanced failure condition The simultaneous occurrence of the crushing of the compression concrete on one side of a member and the yielding of the tensile steel on the other side.

Bond stresses The shear-type stresses produced on the surfaces of reinforcing bars as the concrete tries to slip on those bars.

Camber The construction of a member bent or arched in one direction so that it won't look so bad when the service loads bend it in the opposite direction.

Capacity reduction factors Factors that take into account the uncertainties of material strengths, approximations in analysis, and variations in dimensions and workmanship. They are multiplied by the nominal or theoretical strengths of members to obtain their permissible strengths.

Cast-in-place concrete Concrete cast at the building site in its final position.

Column capital A flaring or enlarging of a column underneath a reinforced concrete slab.

Composite column A concrete column that is reinforced longitudinally with structural steel shapes.

Compression reinforcement Reinforcement added to the compression side of beams to increase moment capacity, increase ductility, decrease long-term deflections, or provide hangars for shear reinforcement.

Concrete A mixture of sand, gravel, crushed rock, or other aggregates held together in a rock-like mass with a paste of cement and water.

Concrete masonry unit (CMU) A masonry unit made of portland cement, water, and mineral aggregates formed into a rectangular prism.

Cover A protective layer of concrete over reinforcing bars to protect them from fire and corrosion.

Cracking moment The bending moment in a member when the concrete tensile stress equals the modulus of rupture and cracks begin to occur.

Creep or plastic flow When a concrete member is subjected to sustained compression loads, it will continue to shorten for years. The shortening that occurs after the initial or instantaneous shortening is called *creep* or *plastic flow* and is caused by the squeezing of water from the pores of the concrete.

Dead load Loads of constant magnitude that remain in one position. Examples: weights of walls, floors, roofs, plumbing, fixtures, structural frames, and so on.

Development length The length of a reinforcing bar needed to anchor or develop its stress at a critical section.

Doubly reinforced beam Concrete beams that have both tensile and compression reinforcing.

Drop panels A thickening of a reinforced concrete slab around a column.

Effective depth The distance from the compression face of a flexural member to the center of gravity of the tensile reinforcing.

Factored load A load that has been multiplied by a load factor, thus providing a safety factor.

Flat plate Solid concrete floor or roof slabs of uniform depths that transfer loads directly to supporting columns without the aid of beams or capitals or drop panels.

Flat slab Reinforced concrete slab with capitals and/or drop panels.

Formwork The mold in which semiliquid concrete is placed.

Grade 40 (60) reinforcement Reinforcement with a minimum yield stress of 40,000 psi (60,000 psi).

Grout A mixture of portland cement or other cement, sand, water, and often coarse aggregate and sometimes admixtures used to fill the hollow cells in masonry units and bond reinforcing to the masonry.

Honeycomb Areas of concrete where there is segregation of the coarse aggregate or rock pockets where the aggregate is not surrounded with mortar. It is caused by the improper handling and placing of the concrete.

Inflection point A point in a flexural member where the bending moment is zero and where the moment is changing from one sign to the other.

Influence line A diagram whose ordinates show the magnitude and character of some function of a structure (shear, moment, etc.) as a load of unity moves across the structure.

Interaction curve A diagram showing the interaction or relationship between two functions of a member, usually axial column load and bending.

Joint reinforcement Wire reinforcing embedded in mortar joints used in masonry walls to reduce shrinkage cracking.

L beam A T beam at the edge of a reinforced concrete slab that has a flange on only one side.

Lightweight concrete Concrete where lightweight aggregate (such as zonolite, expanded shales, sawdust, etc.) is used to replace the coarse and/or fine aggregate.

Limit state A condition at which a structure or some part of that structure ceases to perform its intended function.

Lintel A beam made of structural steel, precast concrete, or masonry embedded in a masonry wall spanning over an opening such as a window or door.

Live loads Loads that change position and magnitude. They move or are moved. Examples: trucks, people, wind, rain, earthquakes, temperature changes, and so on.

Load factor A factor generally larger than one that is multiplied by a service or working load to provide a factor of safety.

Long columns *See* Slender columns.

Maximum considered earthquake (MCE) An extreme earthquake, considered to occur only once every 2500 years.

Microcrack A crack too fine to be seen with the naked eye.

Modulus of elasticity The ratio of stress to strain in elastic materials. The higher its value, the smaller the deformations in a member.

Modulus of rupture The flexural tensile strength of concrete.

Monolithic concrete Concrete cast in one piece or in different operations but with proper construction joints.

Mortar A cementitious material used to construct masonry by being placed between the individual masonry units to bond them together to form a masonry element such as a wall.

Nominal strength The theoretical ultimate strength of a member, such as M_n (nominal moment), V_n (nominal shear), and so on.

One-way slab A slab designed to bend in one direction.

Overreinforced members Members for which the tensile steel will not yield (nor will cracks and deflections appreciably change) before failure, which will be sudden and without warning due to crushing of the compression concrete.

***P-δ* moments** *See* Secondary moments.

Plain concrete Concrete with no reinforcing whatsoever.

Plastic centroid of column The location of the resultant force produced by the steel and the concrete.

Plastic deformation Permanent deformation occurring in a member after its yield stress is reached.

Plastic flow *See* Creep or plastic flow.

Poisson's ratio The lateral expansion or contraction of a member divided by its longitudinal shortening or lengthening when the member is subjected to tension or compression forces. (Average value for concrete is about 0.16.)

Posttensioned concrete Prestressed concrete for which the steel is tensioned after the concrete has hardened.

Precast concrete Concrete cast at a location away from its final position. It may be cast at the building site near its final position, but usually this is done at a concrete yard.

Prestressing The imposition of internal stresses into a structure that are of an opposite character to those that will be caused by the service or working loads.

Pretensioned Prestressed concrete for which the steel is tensioned before the concrete is placed.

Primary moments Computed moments in a structure that do not account for structure deformations.

Ready-mixed concrete Concrete that is mixed at a concrete plant and then is transported to the construction site.

Reinforced concrete A combination of concrete and steel reinforcing wherein the steel provides the tensile strength lacking in the concrete. (The steel reinforcing can also be used to resist compressive forces.)

Secondary moments Moments caused in a structure by its deformations under load. As a column bends laterally, a moment is caused equal to the axial load times

the lateral deformation. It is called a *secondary* or *P-δ moment*.

Seismic Design Category A classification given to a structure based on its Occupancy Category and the severity of the design earthquake ground motion at the site.

Serviceability Pertains to the performance of structures under normal service loads and is concerned with such items as deflections, vibrations, cracking, and slipping.

Service loads The actual loads that are assumed to be applied to a structure when it is in service (also called *working loads*).

Shearheads Cross-shaped elements, such as steel channels and I beams, placed in reinforced concrete slabs above columns to increase their shear strength.

Shearwall A wall usually made of reinforced concrete or reinforced masonry loaded with lateral loads in its own plane.

Shores The temporary members (probably wood or metal) that are used for vertical support for formwork into which fresh concrete is placed.

Short columns Columns with such small slenderness ratios that secondary moments are negligible.

Slender columns (or long columns) Columns with sufficiently large slenderness ratios that secondary moments appreciably weaken them (to the ACI, an appreciable reduction in strength in columns is more than 5%).

Spalling The breaking off or flaking off of a concrete surface.

Spiral column A column that has a helical spiral made from bars or heavy wire wrapped continuously around its longitudinal reinforcing bars.

Spirals Closely spaced wires or bars wrapped in a continuous spiral around the longitudinal bars of a member to hold them in position.

Split-cylinder test A test used to estimate the tensile strength of concrete.

Stirrups Vertical reinforcement added to reinforced concrete beams to increase their shear capacity.

Strength design A method of design whereby the estimated dead and live loads are multiplied by certain load or safety factors. The resulting so-called *factored loads* are used to proportion the members.

T beam A reinforced concrete beam that incorporates a portion of the slab that it supports.

Tendons Wires, strands, cables, or bars used to prestress concrete.

Tied column A column with a series of closed steel ties wrapped around its longitudinal bars to hold them in place.

Ties Individual pieces of wires or bars wrapped at intervals around the longitudinal bars of a member to hold them in position.

Top bars Horizontal reinforcing bars that have at least 12 in. of fresh concrete placed beneath them.

Transformed area The cross-sectional area of one material theoretically changed into an equivalent area of another material by multiplying it by the ratio of the moduli of elasticity of the two materials. For illustration, an area of steel is changed to an equivalent area of concrete, expressed as nA_s.

Two-way slabs Floor or roof slabs supported by columns or walls arranged so that the slab bends in two directions.

Underreinforced member A member that is designed so that the tensile steel will begin to yield (resulting in appreciable deflections and large visible cracks) while the compression concrete has not yet reached its limiting compressive strain. Thus, a warning is provided before failure occurs.

Web reinforcement Shear reinforcement in flexural members.

Working loads The actual loads that are assumed to be applied to a structure when it is in service (also called *service loads*).

Working-stress design A method of design where the members of a structure are so proportioned that the estimated dead and live loads do not cause elastically computed stresses to exceed certain specified values. Method is also referred to as *allowable stress design, elastic design*, or *service load design*.